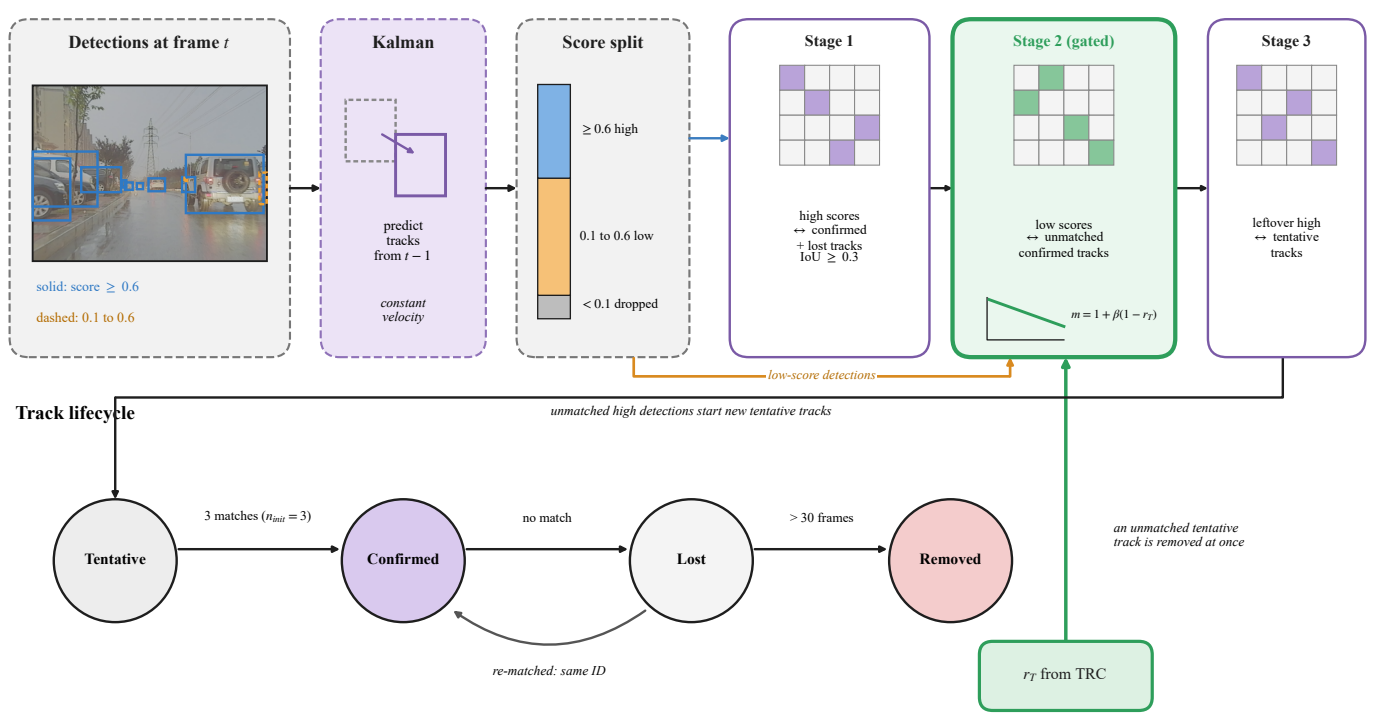

Detections at frame t
solid: score $\geq$ 0.6
dashed: 0.1 to 0.6
Kalman
predict tracks from $t-1$
constant velocity
Score split
$\geq$ 0.6 high
0.1 to 0.6 low
< 0.1 dropped
Stage 1
high scores $\leftrightarrow$ confirmed + lost tracks IoU $\geq$ 0.3
Stage 2 (gated)
low scores $\leftrightarrow$ unmatched confirmed tracks
$m = 1 + \beta(1 - r_T)$
Stage 3
leftover high $\leftrightarrow$ tentative tracks
low-score detections
unmatched high detections start new tentative tracks
Track lifecycle
Tentative
3 matches ($n_{init} = 3$)
Confirmed
no match
Lost
> 30 frames
Removed
re-matched: same ID
an unmatched tentative track is removed at once
r_T from TRC